American Computer Science League

2023-2024 ● Contest 1: Short Problems ● Senior Division

1. Computer Number Systems

If the tickets to an event were numbered sequentially in base 16 and the first one sold was $3A5_{16}$ and the last one sold was $F4A_{16}$, how many tickets were sold expressed in base 10?

A. 2982
B. 2882
C. 2981
D. 2992
E. 2892

2. Computer Number Systems

What is the value of $7245_8 * 17_8$ expressed in base 8?

A. 165553
B. 155653
C. 156653
D. 156553
E. 157553

3. Recursive Functions

Find $f(f(f(15)))$ given:

$$f(x) = \begin{cases} (f(x-2)) \bmod 4 + 2 & \text{if } x \geq 10 \\ (f(x-1)) \bmod 5 + 4 & \text{if } 6 < x < 10 \\ x + 5 & \text{if } x \leq 6 \end{cases}$$

A. 2
B. 4
C. 5
D. 8
E. 11

4. Recursive Functions

Ann decided to make a 96" square quilt**.** The square was divided into 9 equal squares. The center square of each was left alone and the surrounding squares were divided into 9 equal squares. This process continued until the 9 squares formed measured less than 2 inches on a side. How many of the smallest squares were made?

A. 5832
B. 5184
C. 4608
D. 4096
E. 576

5. What Does This Program Do?

What is the output of the following program?

```
for yr = 2021 to 2025
    num = yr
    s1 = 0
    s2 = 0
    while num != 0
        d = num % 10
        s1 = s1 + d
        s2 = s2 + d^2
        num = int(num/10)
    end while
    if yr%s1 == 0 && yr%s2 == 0 then
        output yr
    end if
next yr
```

A. 2021
B. 2022
C. 2023
D. 2024
E. 2025

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1. 计算机数字系统

若某个活动门票以十六进制数按顺序进行编号，第一张出售的门票编号为 $3A5_{16}$，最后一张出售的门票编号为 $F4A_{16}$。请问共卖出多少张门票？答案用十进制数表示。

A. 2982
B. 2882
C. 2981
D. 2992
E. 2892

2. 计算机数字系统

将 $7245_8 * 17_8$ 的值用八进制表示。

A. 165553
B. 155653
C. 156653
D. 156553
E. 157553

3. 递归函数

基于以下表达，求 $f(f(f(15)))$：

$$f(x) = \begin{cases} (f(x-2)) \bmod 4 + 2 & \text{如果 } x \geq 10 \\ (f(x-1)) \bmod 5 + 4 & \text{如果 } 6 < x < 10 \\ x + 5 & \text{如果 } x \leq 6 \end{cases}$$

A. 2
B. 4
C. 5
D. 8
E. 11

4. 递归函数

小安决定做一床 96 英寸的正方形被子。正方形被子可以被分为 9 个相同的小正方形。保持每一块小正方形的中间方格不动，周围方格被分为 9 个相同的更小的正方形。重复操作，直到 9 个正方形的每条边长都小于 2 英寸。求其中最小正方形的总数。

A. 5832
B. 5184
C. 4608
D. 4096
E. 576

5. 这个程序是做什么的？

求下列程序的输出结果。

```
for yr = 2021 to 2025
    num = yr
    s1 = 0
    s2 = 0
    while num != 0
        d = num % 10
        s1 = s1 + d
        s2 = s2 + d^2
        num = int(num/10)
    end while
    if yr%s1 == 0 && yr%s2 == 0 then
        output yr
    end if
next yr
```

A. 2021
B. 2022
C. 2023
D. 2024
E. 2025